

GraphDREAM: Graph-based Decoupled Representation and Evolution-Assisted Multi-task Network for Multimodal Emotion Recognition

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Abstract

Multimodal Emotion Recognition (MER) aims to interpret human emotions from multi-source signals. However, inherent modality disparities cause semantic conflicts and information redundancy during fusion. Moreover, traditional homogeneous graphs fail to distinguish between intra-speaker inertia and external contextual interferences. These issues, further exacerbated by insufficient constraints on emotional logic, lead to affective drift in long dialogues. To address this, we propose GraphDREAM, a multi-task framework following a progressive "purification-topology-calibration" paradigm. First, a Decoupled Cross-modal Refinement (DCR) module decomposes features into shared and private representations to neutralize conflicts and filter noise, extracting purified multimodal features. Building on the purified features, a Spatio-temporal Evolutionary Heterogeneous Graph (SEHG) is developed to disentangle dynamic interferences by explicitly modeling intra-speaker inertia and inter-speaker interactions, thereby effectively fusing these features. Finally, to refine the fused representations, a Multi-task Consistency Calibration (MCC) mechanism is introduced as an auxiliary task, which explicitly characterizes the emotional evolution process via Affective Transition Learning (ATL). By capturing polarity trends, the model effectively models affective dynamics and ensures Evolutionary Coherence in sequential predictions. Extensive experiments on multiple benchmark datasets show that GraphDREAM achieves competitive results against strong baselines. The source code is available at: <https://github.com/Nymph1987/GraphDREAM>.

Keywords: multimodal emotion recognition, decoupled representation learning, heterogeneous graph neural networks, multi-task learning

1 Introduction

Multimodal emotion recognition focuses on modeling human emotions by integrating contextual information, speaker characteristics, and multimodal cues [1–3]. It has important applications in human-computer interaction [4], intelligent education [5], and assisted healthcare [6]. Existing methods typically formulate MER as cross-modal feature fusion and classification problems [7]. Although these approaches have achieved significant progress in processing short dialogue segments [8, 9], they still face fundamental bottlenecks when dealing with complex, dynamic real-world interaction scenarios.

First, direct fusion of heterogeneous signals induces feature entanglement in the joint latent space, leading to cross-modal noise accumulation and degraded emotion discriminability. Conventional fusion paradigms (e.g., simple concatenation or unconstrained attention) lack structural constraints, making it difficult to separate modality-invariant consensus information from modality-specific residuals [10, 11]. As a result, discriminative emotional cues are easily overwhelmed, highlighting the necessity of representation disentanglement for effective fusion. Second, modeling complex topological dependencies in multi-turn dialogues remains challenging. Early graph neural network approaches

[7, 12] rely on homogeneous structures with uniform message passing, which fail to distinguish different relation types. This indiscriminate propagation leads to over-smoothing, blurring the distinction between intra-speaker emotional inertia and inter-speaker interactions. Finally, existing methods often overlook the coherence between discrete emotion classification and continuous emotional evolution. Single-step classification lacks temporal constraints, despite the inherently dynamic nature of emotions. This mismatch makes models prone to affective drift [13], resulting in abrupt and non-smooth transitions in predictions over time. Incorporating emotional evolution modeling with consistency constraints helps improve temporal coherence while maintaining classification performance.

To address these limitations, we propose GraphDREAM, a multi-task network centered on Decoupled Representation and Evolution-Assisted Learning. Unlike existing fusion-based paradigms [7, 12], our model adopts a progressive *purification-topology-calibration* framework to systematically resolve the aforementioned challenges. Initially, to address feature entanglement, we introduce the Decoupled Cross-modal Refinement (DCR) module [14, 15]. Instead of raw fusion, DCR enforces strict orthogonality constraints to explicitly disentangle shared semantics from private residuals, thereby neutralizing cross-modal noise and ensuring that only purified representations enter the reasoning phase. Next, we construct the Spatio-temporal Evolutionary Heterogeneous Graph (SEHG) to overcome the over-smoothing limitations of homogeneous models. By integrating speaker-aware relational edges and a sliding temporal window mechanism, SEHG explicitly delineates intra-speaker emotional inertia pathways from inter-speaker influence channels, enabling fine-grained topological modeling. Ultimately, a Multi-task Consistency Calibration (MCC) mechanism is employed to bridge the gap between discrete predictions and continuous emotional flow. Driven by Affective Transition Learning (ATL), MCC utilizes sentiment polarity shifts and coarse-grained sentiment auxiliary tasks to penalize illogical emotional leaps, thereby imposing a robust evolutionary constraint on the final predictions. Our contributions are summarized as follows:

- 1) **We propose the GraphDREAM framework**, a progressive multi-task architecture that hierarchically addresses modality entanglement, topological over-smoothing, and affective drift through a tightly coupled *purification-topology-calibration* pipeline.
- 2) **We introduce a Decoupled Cross-modal Refinement module** that applies rigorous orthogonality constraints to disentangle shared semantics from modality-specific residuals, deeply purifying representations prior to the graph reasoning stage.
- 3) **We design a Spatio-temporal Evolutionary Heterogeneous Graph** that effectively distinguishes intra-speaker affective inertia from inter-speaker social interference by explicitly encoding speaker-aware boundaries and non-linear dialog topology.
- 4) **We propose a Multi-task Consistency Calibration framework** empowered by Affective Transition Learning, which imposes deep evolutionary constraints via adjacent polarity shifts and coarse-grained sentiment signals to explicitly model emotional evolution trajectories and mitigate sequence prediction drift.
- 5) **We validate the effectiveness of GraphDREAM** through extensive experiments on four benchmark datasets, including IEMOCAP, MELD, MOSEI, and CH-SIMSv2, where it achieves consistently competitive results and clear gains over several strong baselines.

The rest of this paper is organized as follows. Section 2 reviews the related work on multimodal emotion recognition, disentangled representation learning, heterogeneous graph neural networks, and multi-task learning. Section 3 details the proposed GraphDREAM architecture. Section 4 presents the experimental setup, comparative analysis, and ablation studies conducted on several benchmark datasets. Finally, Section 5 concludes the paper, states the current limitations explicitly, and outlines future research directions.

2 Related Work

2.1 Multimodal Emotion Recognition

Multimodal emotion recognition aims to infer fine-grained affective states by integrating textual, acoustic, and visual cues. Early fusion-based methods mainly aggregated heterogeneous modalities through direct concatenation or tensor-style interaction modeling. TFN [16] explicitly captures unimodal and cross-modal interactions through outer-product fusion, while LMF [17] improves efficiency by factorizing the tensor fusion process. These methods establish the basic paradigm of multimodal interaction modeling, but they often treat all cross-modal information as uniformly beneficial and are therefore sensitive to redundancy and modality conflict in realistic conversations.

Later studies introduced stronger interaction modules and context-aware architectures. MulT [18] employs directional cross-modal attention to translate information across modalities and has become a representative interaction baseline. DialogueRNN [9] models emotion flow through speaker-aware recurrent states, whereas DialogueCRN [19] enhances contextual reasoning via cognitive-inspired memory retrieval. More recent conversation MER models, such as JOYFUL [20], HiMul-LGG [21], and GraphSmile [22], combine finer interaction modules with conversational context modeling and achieve stronger empirical performance. In a related multimodal understanding task, MPF

proposes a multi-level perceiving framework that captures hierarchical cross-modal cues, again highlighting the value of structured multimodal interaction design [23].

Despite this progress, interaction-centered MER methods still face a common limitation in conversation scenarios: they usually emphasize feature aggregation, but rarely distinguish transferable shared semantics from modality-private residual interference before deeper contextual reasoning.

2.2 Disentangled Representation Learning

Another line of work therefore focuses on disentangled or constraint-guided multimodal representation learning to suppress modality-specific noise and semantic entanglement. MISA [24] decomposes each modality into shared and private subspaces with orthogonality-related constraints, thereby improving the separation between common semantics and modality-specific residuals. More recent extensions further strengthen this direction with auxiliary consistency objectives and self-supervised regularization to improve robustness under noisy multimodal alignment [25]. These methods substantially improve feature quality, but most of them operate mainly at the representation level. Once the refined features are projected into a shared latent space, residual cross-modal interference and implicit structural dependencies may still persist, especially in emotionally complex dialogues where different modalities exhibit inconsistent reliability. In other words, disentanglement alone does not explicitly address how purified features should be propagated over speaker-aware conversational structure or how prediction trajectories should be calibrated over time. Compared with these methods, our DCR is not introduced as an isolated feature regularizer. Instead, it serves as the purification stage of a larger progressive framework, where shared/private disentanglement is directly coupled with subsequent heterogeneous graph reasoning and temporal calibration.

2.3 Heterogeneous Graph Neural Networks

Graph neural networks have become a mainstream tool for modeling conversational affect because they naturally encode speaker dependencies, contextual turns, and non-Euclidean interaction structures. DialogueGCN [12] represents each utterance as a node and propagates information along speaker-aware temporal edges, establishing a strong graph-based baseline for emotion recognition in conversations. MMGCN [7] extends this idea to multimodal settings by jointly modeling intra-modal and inter-modal connections, thereby improving contextual aggregation across heterogeneous cues. These methods show that explicit relational reasoning is often more effective than purely sequential modeling for multi-party dialogues.

Recent graph-based studies further distinguish relation types to better reflect fine-grained conversational structure. Speaker-aware heterogeneous graph designs [26, 27] model different conversational dependencies with relation-specific parameters, reducing the risk of indiscriminate neighborhood smoothing. Prototype- or structure-guided graph models also improve representation discrimination by regularizing class geometry. For example, the recent Graph-Prototype Distillation framework introduces prototype-guided contrastive supervision for multimodal emotion recognition in conversations, showing the benefit of coupling relational reasoning with more structured representation targets [28]. These advances suggest that graph reasoning becomes more effective when topology design and feature geometry are jointly considered. However, several limitations remain. Many graph models still adopt fully connected or weakly selective interaction patterns, which may blur the boundary between intra-speaker affective inertia and inter-speaker disturbance. In addition, similarity-based dense aggregation can over-smooth emotionally distinct utterances and amplify unreliable signals when the underlying multimodal features are noisy.

2.4 Multi-task Learning

Besides graph topology, multi-task learning has also been explored to stabilize affective prediction by introducing auxiliary sentiment, transition, or consistency signals across utterances [13, 29, 30]. These methods highlight the value of auxiliary supervision for capturing emotion dynamics, but they often rely on sequential modeling alone and do not explicitly combine evolution-aware calibration with heterogeneous graph reasoning. As a result, abrupt local prediction jumps may still occur even when auxiliary supervision is introduced. Compared with these methods, our MCC integrates transition-aware multi-task supervision with graph-based contextual reasoning, so that temporal calibration is applied on top of speaker-aware structured representations rather than in isolation.

Taken together, these four lines of related work motivate the overall design of GraphDREAM: DCR addresses representation purification, SEHG performs heterogeneous topological reasoning, and MCC introduces evolution-aware multi-task calibration. From this perspective, GraphDREAM can be understood as a progressive framework that couples disentangled representation learning, heterogeneous graph reasoning, and auxiliary temporal supervision in a coordinated pipeline.

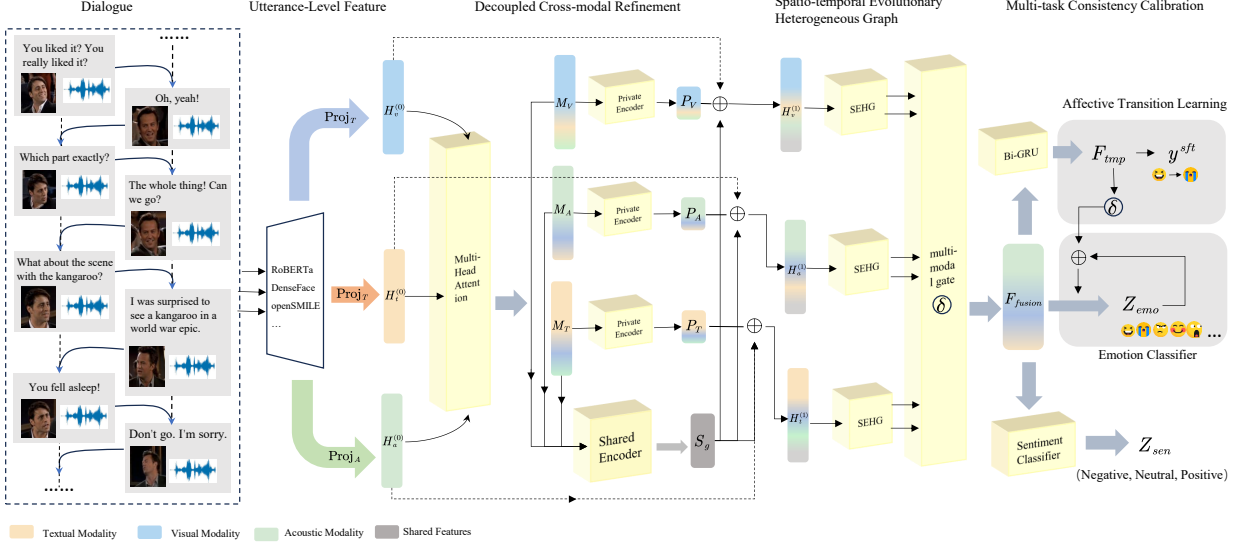


Fig. 1 Overall framework of GraphDREAM, including DCR for shared/private representation refinement, SEHG for speaker-aware spatio-temporal graph propagation, and MCC for ATL-guided temporal calibration.

3 Proposed Method

In this Section, we present the architectural design and implementation details of GraphDREAM. We first establish the problem definition and notations in Section 3.1. As illustrated in Fig. 1, the multimodal emotion recognition task is decomposed into a progressive modeling process consisting of representation purification, evolutionary topology, and sequential calibration. Specifically, the execution of the algorithm unfolds in three critical stages:

First, in the representation purification stage, the model moves beyond the conventional paradigm of processing raw features directly. Instead, it employs DCR to disentangle mixed cross-modal interactions into independent semantic spaces, thereby extracting high-quality node representations for subsequent graph computations; the implementation details of this stage are provided in Section 3.2. Subsequently, the process enters the topological modeling stage elaborated in Section 3.3, where we utilize the SEHG to re-associate static utterance nodes. The graph employs intra-utterance cross-modal connections to model the emotional dependencies between modalities and capture speaker interactions. Additionally, inter-utterance cross-modal connections between adjacent sentences capture the variation of emotional dynamics and their temporal dependencies, focusing on the evolutionary characteristics of affective states within sliding temporal windows. Finally, in the sequential calibration stage described in Section 3.4, we leverage MCC to capture polarity transition patterns between adjacent utterances, rectifying discrete predictions into continuous and smooth emotional trajectories. The detailed mathematical formulations, interaction logic, and the joint optimization objective are unfolded throughout the following subsections, with the final loss function explicitly defined in Section 3.5.

3.1 Problem Definition

Multimodal Emotion Recognition models a dialogue $U = \{u_i\}_{i=1}^B$ to infer the emotion of each utterance. For each utterance u_i , the multimodal input is denoted as $x_i = \{x_i^t, x_i^v, x_i^a\}$, where the textual branch contains multiple views $\{x_i^{t,(k)}\}_{k \in \Omega_t}$. The goal is to predict the fine-grained emotion label $y_{emo} \in \mathcal{C}$ while preserving the dialogue topology induced by speaker identities and temporal masks.

To address these challenges, we propose GraphDREAM, a progressive framework composed of DCR, SEHG, and MCC.

3.2 Decoupled Cross-modal Refinement

Existing decoupling methods typically separate raw unimodal features into shared and private spaces [14, 15]. However, raw signals often lack the cross-modal context necessary for complex emotion modeling. To address this, we propose the DCR module, which performs decoupling after initial interaction to purify high-order representations.

We first project the multimodal inputs into a common hidden space. The textual branch aggregates multiple views $\{x_i^{t,(k)}\}_{k \in \Omega_t}$, and the projected features are denoted as $h_m^{(0)}$, where $m \in \{t, v, a\}$. For each modality m , cross-modal

multi-head attention (MHA) extracts complementary cues from the remaining modalities n and r :

$$\begin{cases} A_{m \leftarrow n} = \text{MHA}(h_m^{(0)}, h_n^{(0)}, h_n^{(0)}) \\ A_{m \leftarrow r} = \text{MHA}(h_m^{(0)}, h_r^{(0)}, h_r^{(0)}) \end{cases} \quad (1)$$

$$M_m = \phi_m([h_m^{(0)}; A_{m \leftarrow n}; A_{m \leftarrow r}]) \quad (2)$$

where M_m represents the interaction-augmented feature.

Unlike conventional paradigms, DCR is applied to M_m to disentangle the cross-modal commonality from the modality-specific residual within the interaction-augmented space. The rationale is that while MHA captures essential correlations, it also introduces cross-modal redundancy and noise that can obscure modality-specific emotional cues. Thus, P_m here represents the purified modality-specific essence refined by cross-modal alignment, rather than raw signal.

We introduce a shared encoder $f_s(\cdot)$ and modality-specific encoders $f_m(\cdot)$. The global shared representation and modality-specific residuals are defined as $\mathcal{L}_{dec}$:

$$S_g = \sum_m \omega_m f_s(M_m), \quad P_m = f_m(M_m) \quad (3)$$

where $\omega_m = \text{Softmax}(\text{MLP}(f_s(M_m)))$ are adaptive weights. To ensure effective separation, we employ a decoupling objective:

$$\mathcal{L}_{dec} = \beta_{ortho} \mathcal{L}_{ortho} + \beta_{align} \mathcal{L}_{align} + \beta_{sep} \mathcal{L}_{sep} \quad (4)$$

Specifically, $\mathcal{L}_{align}$ minimizes the distance between $f_s(M_m)$ across modalities to encourage shared semantics. $\mathcal{L}_{ortho}$ enforces orthogonality between the shared representation S_g and each private representation P_m to reduce information redundancy. $\mathcal{L}_{sep}$ utilizes a similarity constraint to prevent the leakage of unique modality-specific features into the shared space.

The refined representation is reconstructed as:

$$h_m^{(1)} = \text{LN}(h_m^{(0)} + O_m([P_m; S_g])) \quad (5)$$

where O_m is a filtered re-projection. By using the initial $h_m^{(0)}$ as a residual anchor, DCR effectively serves as a pre-propagation filter. It ensures that the subsequent SEHG receives purified features that retain inherent modality characteristics while benefiting from the distilled commonality of cross-modal interactions.

3.3 Spatio-temporal Evolutionary Heterogeneous Graph

While DCR purifies cross-modal representations, it does not model the complex speaker interactions and temporal dependencies in multi-turn dialogues. To capture these, we construct the SEHG, a speaker-aware structure composed of three bi-modal graphs $G_{xy} = (V_{xy}, \mathcal{E}_{xy})$, where $V_{xy} = V_x \cup V_y$ and $x \neq y$. Here, the term evolutionary is realized through sliding temporal windows that propagate affective states along the conversational timeline, enabling the graph to capture the sequential progression of emotional dynamics rather than treating utterances as static, isolated nodes. By restricting message passing within local temporal windows and distinct speaker relations, SEHG prevents global node uniformization, thereby addressing the over-smoothing bottleneck. The overall architecture of SEHG, including its graph construction and propagation mechanism, is illustrated in Fig. 2.

1) Edge Construction: The edge set $\mathcal{E}_{xy}$ is designed to capture speaker-aware interactions within local temporal neighborhoods. For nodes $u_i, u_j \in V_{xy}$, an edge e_{ij} exists only if:

$$e_{ij} \in \mathcal{E}_{xy} \quad \text{iff} \quad (j \in [i - w_p, i + w_f]) \quad (6)$$

where $w_p, w_f \in \mathbb{N}^+$ are integer temporal radii (past and future windows, respectively) and $sp(\cdot)$ denotes speaker identity. In other words, all edges are first restricted to the local window $[i - w_p, i + w_f]$, and are then assigned relation labels according to whether the two utterances come from the same speaker or not. This yields window-sparse local neighborhoods rather than unconstrained dense connectivity. In our experiments, we set $w_p = w_f = 11$. Since our setting follows the standard offline MERC protocol rather than causal online forecasting, both preceding and subsequent local utterances are available at evaluation time; therefore, the symmetric window does not introduce test-time future-label leakage under the benchmark setting. As shown by the extended sensitivity analysis in Section 4.4, IEMOCAP reaches its best performance at $W = 11$ while MELD peaks at $W = 9$; nevertheless, $W = 11$ remains consistently near-optimal across datasets and therefore serves as a stable unified trade-off between sufficient context coverage and noise accumulation from overly broad neighborhoods.

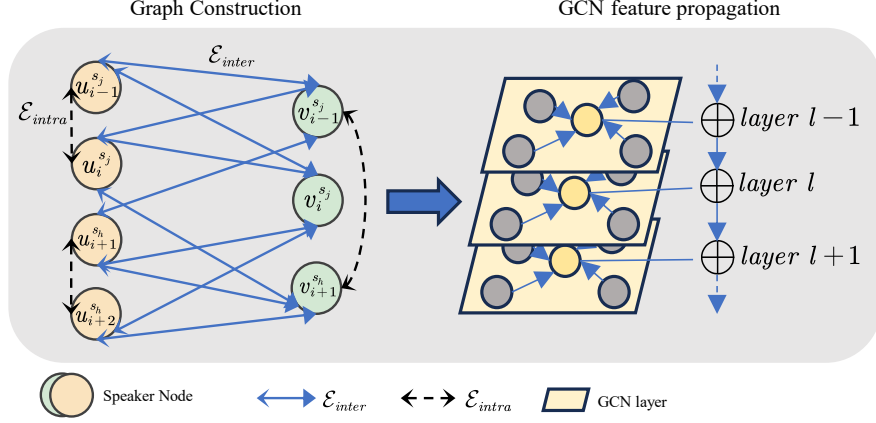


Fig. 2 The architecture of the SEHG. The left panel illustrates the graph construction, where utterance nodes are connected only within a local temporal window and then labeled as inter-speaker or intra-speaker relations according to speaker identity. The right panel depicts the multi-layer GCN used for evolutionary feature propagation.

2) Heterogeneous Propagation and Gating: To differentiate temporal and speaker relations, we adopt a relation-specific propagation mechanism based on a multi-layer Graph Convolutional Network (GCN) with $\mathcal{R} = \{\text{inter}, \text{intra}\}$. For L layers ($L = 6$), the update rule is

$$H_{xy}^{(\ell)} = \text{LN} \left(H_{xy}^{(\ell-1)} + \sigma \left(\sum_{r \in \mathcal{R}} \gamma_r^{(\ell-1)} \hat{A}_{xy,r} H_{xy}^{(\ell-1)} W_r^{(\ell-1)} \right) \right) \quad (7)$$

where $\hat{A}_{xy,r}$, $W_r^{(\ell-1)}$, and γ_r are the normalized adjacency, relation weight, and importance coefficient, respectively. Although γ_r is not an explicit decay kernel, it acts as a relation-level sparse filter that suppresses weak long-range messages; together with residual propagation, this alleviates over-smoothing even under a broad temporal window. We intentionally retain only speaker-aware static edges as the default design rather than adding similarity-derived dynamic implicit edges at this stage, because noisy early representations may create spurious shortcuts between emotionally ambiguous but conversationally unrelated utterances. The current design therefore prioritizes structural interpretability and topological stability. For completeness, we report an exploratory pilot comparison with dynamic-edge variants later in Section 4.4, but they are not adopted in the final model.

The resulting embeddings are divided into six directional branches, namely $\{T \leftarrow V, V \leftarrow T, T \leftarrow A, A \leftarrow T, V \leftarrow A, A \leftarrow V\}$, and fused by dynamic gating:

$$F = \sum_{i=1}^6 \alpha_i \odot B_i, \quad \alpha = \text{Softmax}(\text{MLP}([B_1; \dots; B_6])) \quad (8)$$

where $\alpha \in \mathbb{R}^6$ are learnable weights.

3.4 Multi-task Consistency Calibration

Even with SEHG, discrete emotion prediction from aligned representations can still suffer from affective drift—inconsistent jumps between adjacent utterances that violate natural emotional evolution[31]. To enforce temporal smoothness, MCC first processes F with a BiGRU to capture sequential context F_{imp} , and then generates the calibrated feature $\hat{F}$ via a gated residual mechanism:

$$\hat{F} = F + \sigma(W_g[F; F_{imp}] + b_g) \odot \tanh(F_{imp}) \quad (9)$$

where $\hat{F}$ is utilized for emotion inference. To improve class separability, we introduce a supervised contrastive loss. Let $z_i = \text{MLP}(\hat{F}_i) / \|\text{MLP}(\hat{F}_i)\|_2$ be the projected representation for sample i in batch I :

$$\mathcal{L}_{con} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i \cdot z_p / \tau)}{\sum_{a \in I \setminus \{i\}} \exp(z_i \cdot z_a / \tau)} \quad (10)$$

where $P(i)$ is the positive set for anchor i and τ is the temperature. Building upon this, we further bridge discrete labels and continuous evolution through two auxiliary tasks:

1) Sentiment Mapping via Polarity Mapping: While fine-grained emotional labels exhibit high volatility in long dialogues, coarse-grained sentiment polarities (*Positive, Neutral, Negative*) maintain higher stability. We define a mapping function $\mathcal{M} : \mathcal{C} \rightarrow \{\text{pos, neu, neg}\}$ to derive the polarity label y_{sen} from y_{emo} . This mapping enables the model to learn high-level semantic constraints, leveraging broad sentiment trends to suppress localized emotional noise.

2) Affective Transition Learning and Consistency Constraints: To explicitly model the evolutionary laws of emotions, we introduce ATL, which focuses on the polarity evolution between adjacent utterances. In the original training setting, ATL adopts a teacher-forcing-style target construction, namely $y_{tr,i} = \mathcal{T}(y_{sen,i-1}, y_{sen,i})$. Importantly, the current polarity label $y_{sen,i}$ is used only after the forward pass to define the supervision target of the auxiliary transition branch, rather than being fed into the multimodal encoder or the main emotion classifier as an input feature. To prevent over-smoothing while ensuring logical progression, ATL works in synergy with the temporal-consistency loss $\mathcal{L}_{tc}$:

$$\mathcal{L}_{tc} = \sum_i \|\hat{F}_i - \hat{F}_{i-1}\|_2^2 \quad (11)$$

where $\mathcal{L}_{tc}$ ensures smooth transitions by constraining the Euclidean distance between adjacent features, while $\mathcal{L}_{atl}$ identifies critical emotional shifts within these transitions. During inference, the model has no access to any ground-truth polarity labels and relies solely on observed multimodal inputs and historical context. We further evaluate a strict-history ATL variant in Section 4.4, where the auxiliary transition target is constructed only from history-derived predicted polarity proxies, to verify that the main conclusion does not rely on inference-time access to future gold labels.

By integrating constrained smoothness with transition awareness, MCC ensures that the calibrated representations preserve both emotional discriminability and evolutionary continuity.

3.5 Joint Optimization

The DCR, SEHG, and MCC modules are trained in an end-to-end manner. The overall objective integrates task-level supervision with a small set of representation regularizers:

$$\begin{aligned} \mathcal{L} = & \alpha_{emo} \mathcal{L}_{emo} + \alpha_{sen} \mathcal{L}_{sen} + \alpha_{atl} \mathcal{L}_{atl} \\ & + \alpha_{dec} \mathcal{L}_{dec} + \alpha_{con} \mathcal{L}_{con} + \alpha_{tc} \mathcal{L}_{tc} + \alpha_{aux} \mathcal{L}_{aux} \end{aligned} \quad (12)$$

where $\mathcal{L}_{emo}$, $\mathcal{L}_{sen}$, and $\mathcal{L}_{atl}$ denote the emotion, sentiment, and ATL losses, respectively; $\mathcal{L}_{dec}$ is the decoupling loss; $\mathcal{L}_{con}$ denotes the contrastive regularization term; and $\mathcal{L}_{tc}$ is the temporal-consistency loss defined in Section 3.4. In addition, $\mathcal{L}_{aux}$ summarizes several lightweight auxiliary regularization terms used to improve fine-grained emotion discrimination and prediction consistency, without introducing extra inference branches. Typical examples include a pair-wise similarity penalty and a prototype-aware compactness regularizer, together with other lightweight training-only stabilizers. The weights α are tuned on the validation split.

GraphDREAM is designed as a coordinated system rather than a collection of isolated components. By passing the refined representations from DCR into the spatio-temporal topology of SEHG, the framework transforms static multimodal inputs into a feature stream that reflects conversational evolution more explicitly. MCC then further regularizes this representation with auxiliary temporal supervision, which helps reduce abrupt local prediction jumps and improves temporal coherence. In this way, the overall framework couples purification, topological reasoning, and calibration in a single progressive pipeline for multimodal emotion recognition.

4 Experimental Results and Analysis

In this Section, we present the experimental setup, evaluate the performance of our model on several benchmark datasets, and conduct a series of ablation studies to verify the effectiveness of each module.

4.1 Datasets and Metrics

To evaluate the robustness of GraphDREAM, we conduct experiments on four benchmark datasets spanning diverse conversational scenarios and linguistic backgrounds. These include IEMOCAP [32], a controlled dyadic dataset featuring high-quality emotional interactions between professional actors; MELD [33], derived from the sitcom *Friends* with multi-speaker interactions and complex unstructured background noise; CMU-MOSEI [34], the largest monologue collection posing challenges in data imbalance and speaker diversity; and CH-SIMSV2 [35], which provides multi-grained independent annotations within the context of Chinese films and TV dramas. The detailed distribution of conversations and utterances is summarized in Table 1. Furthermore, as illustrated in Fig. 3, these datasets exhibit

significant class imbalance (e.g., the high proportion of ‘Fear’ and ‘Disgust’ in MELD), which necessitates robust modeling of subtle affective transitions.

Table 1 Statistics of conversation and utterance distributions across the training, validation, and testing sets for the benchmark datasets.

Dataset	Conversation			Utterance		
	Train	Val	Test	Train	Val	Test
IEMOCAP	108	12	31	5163	647	1623
MELD	1038	114	280	10037	1061	2610
CMU-MOSEI	2249	300	676	16326	1871	4659
CH-SIMSV2	129	120	142	2722	647	1034

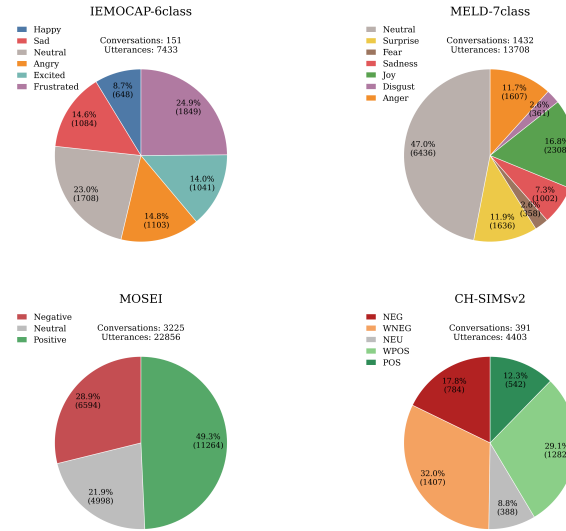


Fig. 3 Class distribution across the four benchmark datasets. The percentage and raw counts highlight the inherent category imbalance.

To support the auxiliary tasks in our MCC module (Section 3.4), we implement a mapping function $\mathcal{M}$ to group the original fine-grained labels into three sentiment polarities: *Negative*, *Neutral*, and *Positive*, as detailed in Table 2. Notably, when training for binary sentiment tasks (e.g., ACC2), we utilize this tripartite mapping to perform auxiliary three-class supervision rather than simply discarding neutral samples. By retaining the *Neutral* category during training, the model learns more robust emotional boundaries, providing stable semantic constraints for the primary binary classification.

Table 2 Mapping of fine-grained emotion labels into three sentiment polarities (Negative, Neutral, Positive) across different datasets.

Sentiment	IEMOCAP	MELD	CMU-MOSEI	CH-SIMSV2
Negative	Sad, Angry, Frustrated	Sadness, Fear, Disgust, Anger	Negative	Negative, Weakly Negative
Neutral	Neutral	Neutral	Neutral	Neutral
Positive	Happy, Excited	Joy, Surprise	Positive	Weakly Positive, Positive

Table 3 Six-run statistical significance results on IEMOCAP and MELD. Run1–Run6 correspond to random seeds 1010, 123, 42, 456, 789, and 2024, respectively. We report the corpus-level Acc and weighted F1 for each run, together with mean $\pm$ standard deviation. The paired t-test is conducted on weighted F1 between GraphDREAM and GraphSmile using the same seeds for run pairing.

Dataset	Model	Metric	Run1	Run2	Run3	Run4	Run5	Run6	Mean $\pm$ Std	Paired p -value
IEMOCAP	GraphSmile	Acc	69.84	70.24	69.92	69.93	69.95	69.99	69.98 $\pm$ 0.14	–
		F1(w)	69.58	70.00	69.87	70.02	69.85	69.98	69.88 $\pm$ 0.16	–
	GraphDREAM	Acc	72.43	72.32	72.38	72.75	72.44	72.46	72.46 $\pm$ 0.15	–
		F1(w)	72.45	72.03	72.41	72.30	72.68	72.52	72.40 $\pm$ 0.22	$p < 0.001$
MELD	GraphSmile	Acc	66.25	66.51	66.47	66.56	66.45	66.44	66.45 $\pm$ 0.11	–
		F1(w)	64.52	63.96	64.32	64.53	64.87	64.45	64.44 $\pm$ 0.30	–
	GraphDREAM	Acc	67.05	66.21	66.28	66.09	66.86	66.55	66.51 $\pm$ 0.38	–
		F1(w)	65.98	66.02	65.96	66.44	65.77	66.08	66.04 $\pm$ 0.22	$p < 0.001$

Following standard protocols, we employ Accuracy (Acc) and weighted F1-score (F1) as the primary evaluation metrics. Both corpus-level results and class-wise scores for specific datasets are reported to provide a comprehensive performance analysis.

4.2 Implementation Details

GraphDREAM is implemented in PyTorch 2.7.1 and trained on a Linux server equipped with NVIDIA GPUs. The model is optimized using the AdamW optimizer with the AMSGrad variant, with an initial learning rate of 8×10^{-5} , a weight decay of 1×10^{-4} , and a batch size of 16. To ensure reproducibility and fair comparison, the input feature dimensions for the Text/Audio/Visual modalities are strictly aligned with prior works: 1024/1582/342 for IEMOCAP, 1024/300/342 for MELD, 768/74/35 for MOSEI, and 768/25/177 for CH-SIMSv2. Across all benchmarks, the textual modality adopts 1024-D or 768-D embeddings, denoted as text0–text3.

The model employs a hidden dimension of 384 with 4 attention heads in the DCR module. The SEHG encoder is configured with symmetric temporal windows ($w_p = w_f = 11$) and a 6-layer heterogeneous graph convolution backbone. Regularization includes a dropout rate of 0.28 and a graph residual dropout of 0.15. The overall objective is a joint loss function integrating emotion classification, sentiment supervision, contrastive learning, decoupling regularization, and temporal consistency. Specifically, for the IEMOCAP dataset, the loss weights are set as $(\alpha_{emo}, \alpha_{sen}, \alpha_{atl}) = (1.22, 0.68, 0.40)$, $\alpha_{dec} = 0.51$, $\alpha_{con} = 0.032$, and $\alpha_{tc} = 0.11$. The auxiliary loss $\mathcal{L}_{aux}$ is instantiated as a weighted combination of lightweight training-only regularizers, including a pair-wise similarity penalty and a prototype-aware compactness regularizer, to improve local discrimination and prediction stability.

4.3 Comparison With Representative Baselines

To evaluate the effectiveness of GraphDREAM, we compare it against several representative and competitive baselines, including the foundational fusion model LMF [17], the classic contextual modeling approach DialogueGCN [12], the self-supervised robustness-enhanced framework Self-MM [36], and the strong recent baseline GraphSmile [22].

To further assess statistical reliability, we repeated GraphSmile and GraphDREAM over six random seeds on the two main conversation benchmarks and summarize the per-seed corpus-level results in Table 3. Using paired t-tests on weighted F1, which is the primary evaluation metric in this paper, and pairing runs with the same random seeds across the two models, we observe that GraphDREAM significantly outperforms GraphSmile on both IEMOCAP and MELD, with $p < 0.001$ in both cases. These results indicate that the reported gains are robust rather than caused by random fluctuations.

Compared with a broad spectrum of competitive baselines, GraphDREAM demonstrates consistent gains on the IEMOCAP and MELD datasets, as shown in Table 4. On IEMOCAP, our model achieves a weighted F1-score of 72.40 ± 0.22 , surpassing GraphSmile’s 69.88 ± 0.16 and JOYFUL by 1.37 absolute points. This improvement is particularly evident in the “Excited” 81.65% and “Frustrated” 72.30% categories. The confusion matrix in Fig. 4 further illustrates this: GraphDREAM exhibits higher diagonal values and reduced off-diagonal misclassifications relative to GraphSmile. These gains are consistent with the complementary roles of DCR and SEHG: DCR’s instance-level refinement purifies modality-specific features to filter out sensory noise, while SEHG’s spatio-temporal evolutionary topology captures subtle shifts in speaker-specific states that static methods often overlook. On MELD, GraphDREAM achieves an F1-score of 66.04 ± 0.22 , compared with 64.44 ± 0.30 for GraphSmile. This pattern is consistent with the

Table 4 Performance comparison on IEMOCAP and MELD. For IEMOCAP, the six emotion columns are class-wise scores, while ACC and F1(w) are corpus-level metrics. For MELD, only corpus-level ACC and F1(w) are reported in this table. For GraphSmile and GraphDREAM, the corpus-level metrics are reported as mean $\pm$ standard deviation over six random seeds. The best results are in **bold** and the second-best are underlined.

Models	IEMOCAP-6class						MELD-7class			
	Happy	Sad	Neutral	Angry	Excited	Frustrated	ACC	F1(w)	ACC	F1(w)
DialogueRNN [9]	39.00	83.37	57.26	65.92	74.05	56.23	63.65	63.34	59.35	57.80
DialogueGCN [12]	41.28	82.52	64.33	65.18	73.18	64.46	66.85	66.78	59.58	58.17
DialogueCRN [19]	53.23	83.37	62.96	66.09	75.40	66.07	67.16	67.21	61.11	58.67
MMGCN [7]	37.61	79.84	62.26	<u>74.29</u>	75.00	63.68	67.28	66.67	66.02	64.55
JOYFUL [20]	<u>60.94</u>	<u>84.42</u>	68.24	69.95	73.54	67.55	<u>70.55</u>	<u>71.03</u>	62.53	61.77
COGMEN [37]	51.90	81.70	<u>68.60</u>	66.00	75.30	58.20	68.20	67.60	—	—
GraphCFC [38]	43.08	84.99	64.70	71.35	78.86	63.70	69.13	68.91	61.42	58.86
SACL-LSTM [39]	51.30	82.25	71.39	67.78	75.26	66.94	<u>70.55</u>	70.60	64.52	64.55
MM-DFN [40]	37.61	79.84	62.26	<u>74.29</u>	75.00	63.68	67.28	66.67	61.07	57.33
HiMul-LGG[21]	50.79	83.04	60.25	76.86	77.19	63.90	69.28	69.17	65.29	64.56
GraphSmile [22]	60.98	80.08	62.97	73.33	<u>80.15</u>	66.32	69.98 $\pm$ 0.14	69.88 $\pm$ 0.16	66.45 $\pm$ 0.11	64.44 $\pm$ 0.30
DER-GCN [41]	58.80	79.80	61.50	72.10	73.30	<u>67.80</u>	69.70	69.40	66.85	<u>65.98</u>
GraphDREAM	55.10	81.59	67.95	72.19	81.65	72.30	72.46$\pm$0.15	72.40$\pm$0.22	<u>66.51$\pm$0.38</u>	66.04$\pm$0.22

Table 5 Performance comparison on MOSEI and CH-SIMSV2. For GraphSmile and GraphDREAM, all reported metrics are mean $\pm$ standard deviation over six random seeds. The best results are in **bold** and the second-best are underlined. MOSEI: Has0 (all samples including neutral), Non0 (excluding neutral samples).

Models	CMU-MOSEI					CH-SIMSV2				
	Has0_ACC2	Has0_F1	Non0_ACC2	Non0_F1	AVG	ACC2	ACC3	ACC5	F1-2class	AVG
TFN [16]	78.50	78.96	81.89	81.74	80.27	80.14	72.21	52.55	80.14	71.26
LMF [17]	80.54	80.94	83.48	83.36	82.08	74.18	64.90	47.79	73.88	65.19
Self-MM [36]	83.76	83.82	85.15	84.90	84.41	79.69	72.61	52.77	79.76	71.21
HiMul-LGG [21]	81.35	81.14	83.87	83.69	82.51	79.11	73.89	47.58	79.10	69.92
GraphSmile [22]	83.49 $\pm$ 0.07	82.96 $\pm$ 0.19	85.09 $\pm$ 0.15	84.88 $\pm$ 0.20	84.10 $\pm$ 0.10	78.89 $\pm$ 0.13	<u>74.66$\pm$0.06</u>	51.32 $\pm$ 0.19	78.85 $\pm$ 0.19	70.93 $\pm$ 0.14
EmoE [42]	83.43	83.24	85.50	85.50	84.42	<u>80.85</u>	71.08	53.71	<u>80.76</u>	71.60
TETFN [43]	<u>84.12</u>	<u>84.35</u>	86.21	86.11	<u>85.20</u>	79.73	73.65	<u>54.47</u>	79.81	<u>71.92</u>
CENet [44]	83.52	83.85	<u>86.38</u>	<u>86.32</u>	85.02	79.56	73.10	53.04	79.63	71.33
GraphDREAM	84.59$\pm$0.11	84.43$\pm$0.15	86.43$\pm$0.10	86.43$\pm$0.25	85.47$\pm$0.17	81.33$\pm$0.17	76.00$\pm$0.21	57.68$\pm$0.19	81.26$\pm$0.18	74.07$\pm$0.26

MCC mechanism, which leverages stable sentiment polarity as auxiliary supervision to refine decision boundaries for minority classes and mitigate the impact of severe class imbalance.

On the large-scale CMU-MOSEI and CH-SIMSV2 benchmarks, GraphDREAM also achieves strong results, as detailed in Table 5. For MOSEI, our model achieves an average score of 85.47 ± 0.17 , outperforming GraphSmile’s 84.10 ± 0.10 . This is consistent with DCR’s role in extracting more discriminative representations from unstructured, noisy monologue data by suppressing non-affective redundancies. On CH-SIMSV2, GraphDREAM reaches an ACC5 of 57.68 ± 0.19 , higher than GraphSmile’s 51.32 ± 0.19 and also above TETFN. These improvements support the effectiveness of the “purification-topology-calibration” design under our benchmark settings: SEHG maintains the integrity of long-distance conversational dependencies, while MCC-guided sequential calibration helps keep fine-grained predictions more consistent with the broader sentiment trajectory during rapid emotional transitions.

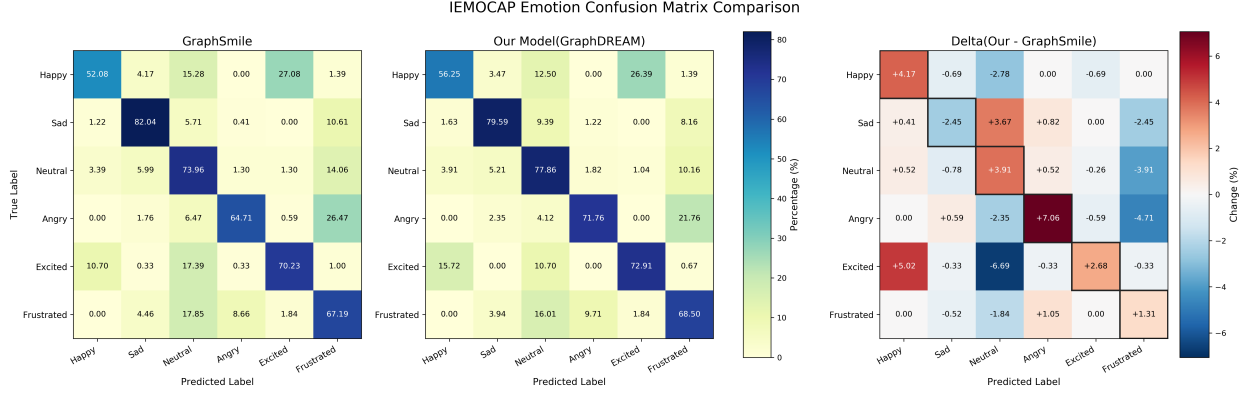


Fig. 4 Confusion matrix comparison between GraphSmile and GraphDREAM on the IEMOCAP dataset. The left and middle panels display normalized distributions for six-class recognition, while the right panel (Delta) highlights the performance gain (red) and reduction in misclassification (blue) of our model.

4.4 Ablation Study

Compared with MOSEI and MELD, which are often constrained by environmental noise or class imbalance, we prioritize IEMOCAP as the primary benchmark for ablation analysis. Its high-fidelity dyadic interactions and fine-grained annotations provide an ideal controlled setting to observe how GraphDREAM handles modality interference and emotional contagion. By systematically decomposing the architecture, we aim to verify the necessity of the proposed *purification-topology-calibration* paradigm in addressing feature entanglement, topological homogenization, and affective drift. The quantitative results are summarized in Table 6.

Table 6 Ablation results on IEMOCAP. Δ denotes the absolute performance drop compared with the Full Model.

Setting	Acc (%)	F1 (%)	Δ Acc	Δ F1
Full Model	72.46	72.52	—	—
<i>Module-level Ablation</i>				
w/o DCR (Representation Purification)	65.56	65.40	-6.90	-7.12
w/o SEHG (Topological Modeling)	68.15	68.04	-4.31	-4.48
w/o MCC (Sequential Calibration)	69.38	69.40	-3.08	-3.12
DCR $\rightarrow$ MulT (Original)	69.98	69.12	-2.48	-3.40
DCR $\rightarrow$ † MulT (Tuned)	71.66	71.13	-0.80	-1.39
<i>Loss & Evolution Ablation</i>				
w/o $\mathcal{L}_{dec}$	70.61	70.67	-1.85	-1.85
w/o $\mathcal{L}_{con}$	71.53	71.56	-0.93	-0.96
w/o $\mathcal{L}_{atl}$	71.04	71.10	-1.42	-1.42
ATL (Strict-history)	70.12	70.37	-2.34	-2.15
w/o $\mathcal{L}_{tc}$	69.62	69.81	-2.84	-2.71
w/o $\mathcal{L}_{atl}$ & $\mathcal{L}_{tc}$	66.42	66.55	-6.04	-5.97
<i>Graph Topology Ablation</i>				
w/o Temporal Edges	69.07	69.17	-3.39	-3.35
w/o Speaker Edges	68.27	68.22	-4.19	-4.30
Homogeneous Graph	68.95	69.01	-3.51	-3.51
w/o Residual	70.98	70.74	-1.48	-1.78
w/o Window	68.45	68.45	-4.01	-4.07

Note: *Homogeneous Graph* preserves the local temporal window and residual propagation, but collapses all relation types into one shared graph relation. *w/o Residual* preserves the heterogeneous speaker / temporal relations and the local window, but removes residual graph propagation. *w/o Window* preserves heterogeneous relations and residual propagation, but removes the local temporal constraint and uses dense inter-utterance connectivity. *w/o Speaker Edges* removes intra-speaker edges only, while *w/o Temporal Edges* removes the temporal inter-utterance edges only. † MulT (tuned) indicates that hyperparameters were searched on the validation set under the same SEHG+MCC pipeline.

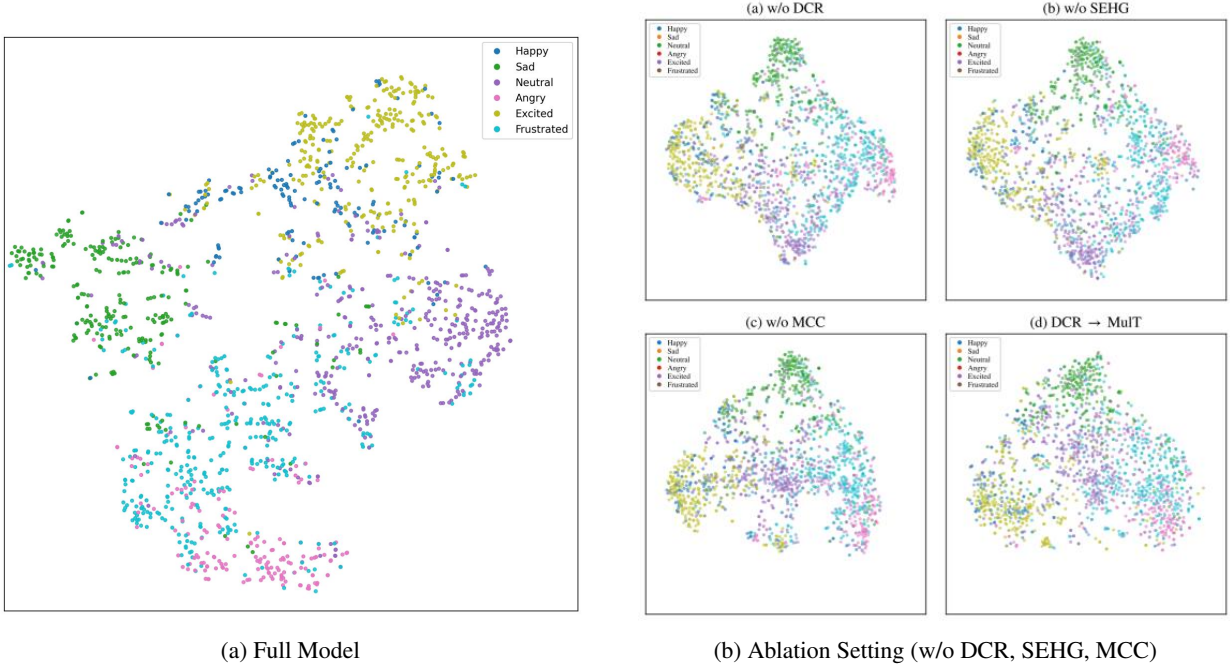


Fig. 5 T-SNE visualization comparison on IEMOCAP. The contrast between (a) and (b) demonstrates that the core modules effectively mitigate semantic conflicts and enhance the discriminative power of the joint affective space.

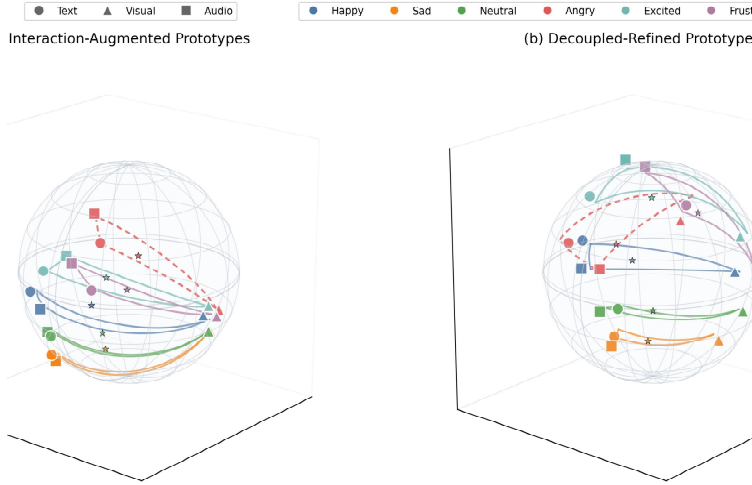


Fig. 6 3D prototype-sphere visualization of multimodal features on IEMOCAP. (a) Interaction-augmented prototypes before decoupled refinement; (b) Decoupled-refined prototypes after DCR. Colors denote emotion classes, marker shapes indicate different modalities (text, visual, and audio), and curved edges connect prototypes of the same emotion across modalities.

To intuitively evaluate these contributions, we first conduct a macro-level visualization of the affective embedding space. As illustrated in the t-SNE comparison Fig. 5, the full GraphDREAM architecture exhibits significantly clearer inter-class boundaries and more compact clusters than the ablation setting (where DCR, SEHG, and MCC are removed). This contrast confirms that our progressive paradigm—moving from representation purification to topological reasoning and sequential calibration—effectively mitigates semantic conflicts and enhances the discriminative power of the joint emotional representations.

1) Representation Purification: How DCR Serves as the Critical Starting Point. As the key initial step in feature purification, the DCR module’s critical role is validated by our ablation study: removing this module results in a 7.12% drop in F1-score, the largest among all modules. To address the fairness concern of a direct replacement, we additionally tuned the replacement [†]MulT on the IEMOCAP validation set under the same SEHG+MCC pipeline. The

best DCR $\rightarrow$ [†]MuT variant (heads = 4, hidden_dim = 384, num_layers = 3) reaches 71.13% F1, which is substantially stronger than the untuned replacement but still clearly below the full model. This confirms that the remaining gap is not simply due to unfair hyperparameter adaptation. To provide direct quantitative evidence for the purification effect itself, we further analyze the refinement-stage representations before graph propagation. Specifically, GraphDREAM yields lower refined average cross-modal cosine similarity than the tuned [†]MuT variant (0.124 vs. 0.141), as well as lower refined mean absolute feature correlation (0.203 vs. 0.256). These two indicators directly reflect redundant cross-modal coupling and feature-wise entanglement, and the lower values of GraphDREAM indicate that DCR suppresses redundant interaction more effectively during cross-modal refinement. Combined with the 3D prototype-sphere in Fig. 6, these results show that DCR does not merely enhance interaction, but explicitly removes redundant shared interference through orthogonality-guided decoupling before graph reasoning. This preemptive purification ensures that the subsequent graph reasoning stage operates on cleaner and less entangled representations, fundamentally enhancing the model’s robustness and accuracy when handling complex emotional patterns.

2) Evolutionary Topology: How SEHG Models Complex Conversational Logic. The ablation result of SEHG (−4.48%) directly supports our claim that homogeneous graphs fail to distinguish relation types. Removing either speaker edges (−4.30%) or temporal edges (−3.35%) causes substantial performance loss, indicating that SEHG explicitly models both intra-speaker emotional inertia and inter-speaker interactions. Moreover, the homogeneous-graph control preserves the same local temporal window and residual propagation, yet still drops by 3.51% in F1, confirming that relation awareness itself is critical rather than being conflated with other implementation details. The newly added control variables further separate the contribution of each component: removing residual propagation leads to a smaller but still clear degradation (−1.78% F1), showing that residual accumulation stabilizes multi-layer graph reasoning, whereas removing the temporal window causes a larger drop (−4.07% F1), indicating that locality constraints are important for preventing dense noisy aggregation. To further examine whether similarity-derived dynamic implicit edges should replace or augment the static topology, we additionally conducted a pilot comparison on IEMOCAP with three settings: Static Only, Dynamic Only, and Static + Dynamic. We emphasize that this comparison is intended as a preliminary pilot study rather than a fully optimized dynamic-graph benchmark; the goal is to test whether naive similarity-based edges provide immediate benefits under the current GraphDREAM configuration. The proposed static topology achieves the best performance (72.46% Acc / 72.52% F1), whereas Dynamic Only drops substantially to 67.71% Acc / 67.87% F1 and Static + Dynamic reaches only 69.99% Acc / 69.96% F1. These results suggest that similarity-derived dynamic edges are not automatically beneficial in MERC: when used alone, they discard the explicit conversational structure encoded by speaker identity and temporal order, and when fused with the static graph they still introduce noisy shortcuts and semantically misleading long-range links that weaken structural selectivity. Therefore, we retain the static speaker-aware topology as the default design under the current benchmark setting because it provides the best balance of conversational faithfulness, interpretability, and empirical robustness, while dynamic implicit edges remain a meaningful future direction that should be coupled with more reliable representation purification and adaptive graph sparsification. Taken together, these controlled results clarify that heterogeneous relation modeling, local window sparsity, and residual propagation each make distinct contributions, with relation awareness and locality being especially important. By separating these factors explicitly, SEHG alleviates the over-smoothing problem caused by indiscriminate message passing in conventional graph-based methods. As illustrated in Fig. 7, this fine-grained evolutionary topology is crucial for capturing the dynamic logic of multi-turn conversations.

3) Sequential Calibration: How MCC Suppresses Affective Drift. As discussed earlier, the absence of logical temporal constraints can cause affective drift in long conversations. The ablation of MCC (−3.12%) and its core supervisory components $\mathcal{L}_{atl}$ and $\mathcal{L}_{tc}$ (−5.97%) confirms the calibration capability of this module. In the original formulation, ATL uses a teacher-forcing-style target construction, but the current-step gold polarity is only used to define the auxiliary supervision target after the forward pass rather than as a model input. To further rule out label leakage concerns, we additionally evaluate a strict-history ATL variant, in which the transition target is constructed only from history-derived predicted polarity proxies instead of directly using the current gold polarity. This stricter control reaches 70.12% Acc and 70.37% F1, compared with 72.46% Acc and 72.52% F1 for the original teacher-forcing ATL. The observed drop indicates that this teacher-forcing-style target construction provides a stronger signal during training, but it does not imply inference-time leakage, because no future gold polarity labels are available at test time. As shown in Fig. 8, MCC effectively smooths unreasonable local jumps in the prediction trajectory, making the final outputs more temporally coherent and logically consistent.

4) Robustness Evaluation: Synergy of Paradigm and Hyperparameter Stability. Fig. 9 presents the impact of key hyperparameters on model performance, while Fig. 10 further extends the window analysis with cross-dataset and dialogue-length-stratified evidence. At the corpus level, IEMOCAP reaches the best performance at $W = 11$ with 72.52 weighted F1, whereas MELD peaks at $W = 9$ with 66.11 weighted F1. Nevertheless, $W = 11$ remains a robust unified operating point: it is optimal on the main benchmark IEMOCAP and remains near-optimal on MELD at 65.88 weighted F1. The additional MELD stratified analysis further shows that the preferred receptive range changes across dialogue-length groups, with $W = 11$ best for short dialogues, $W = 9$ for medium dialogues, and $W = 5$ for long

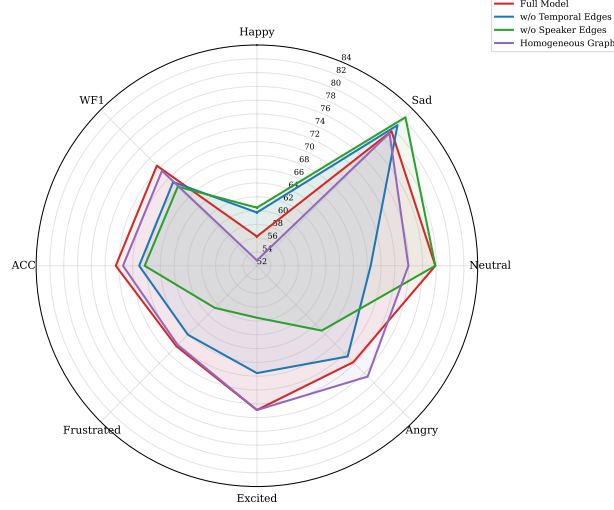


Fig. 7 Radar plot of graph topology, highlighting the performance impact of different edge configurations (homogeneous vs. heterogeneous) within the SEHG module.

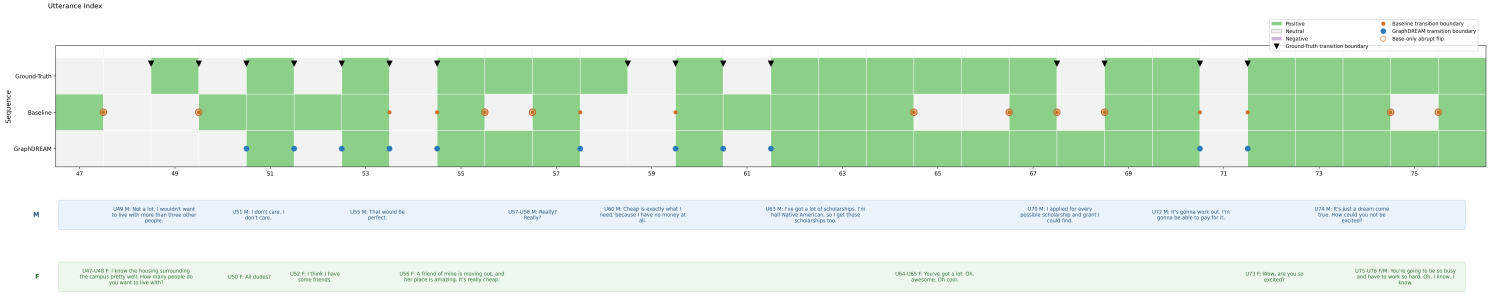


Fig. 8 Discrete emotional polarity evolution over utterance index for a representative IEMOCAP dialogue segment. Ground-Truth (black triangles), Baseline (orange circles), and GraphDREAM (blue circles) are shown, with transition boundaries marked: baseline-only abrupt flips (orange hollow circles), GraphDREAM transition boundaries (solid blue circles), and ground-truth transition boundaries (black solid triangles). The bottom panels display the corresponding speaker turns (M: male, F: female) and utterances.

dialogues, which supports our revised interpretation that a fixed window is a practical trade-off rather than a universally optimal constant. Intuitively, smaller windows fail to cover sufficient emotional carry-over, while overly large windows introduce noisy long-range interactions and dilute relation selectivity. Together with the optimal graph depth $L = 6$, these results demonstrate that the robustness of GraphDREAM stems from the tight coupling among its modules: DCR provides a purified starting point, SEHG serves as a structured reasoning bridge, and MCC delivers a logically calibrated endpoint. An adaptive window mechanism will be explored in future work.

Overall, our ablation studies show that the effectiveness of GraphDREAM comes from the coordination between the progressive pipeline and its multi-task objectives. DCR, together with $\mathcal{L}_{dec}$, establishes a cleaner representation foundation, SEHG provides the structural backbone through heterogeneous topology, and MCC improves logical consistency through transition-aware supervision $\mathcal{L}_{atl}$ and $\mathcal{L}_{tc}$. The consistent performance degradation across the ablation settings supports the usefulness of combining purification, topological reasoning, and calibration for multimodal emotion recognition.

4.5 Case Study

To validate the internal synergy of GraphDREAM, we analyze a monologue transitioning from novelty to frustration, as illustrated in Fig. 11. The mechanism-oriented analysis reveals a three-stage optimization pipeline: first, DCR purifies features by filtering modal noise, which significantly enhances intra-class cohesion as evidenced by the distinct clustering of emotional prototypes (stars vs. circles) in the bottom-left panel; second, SEHG captures long-distance intra-speaker dependencies by aggregating historical emotional inertia, where T4’s high similarity to its historical context (T2/T3) ensures temporal continuity through evolutionary topology; finally, MCC rectifies local fluctuations

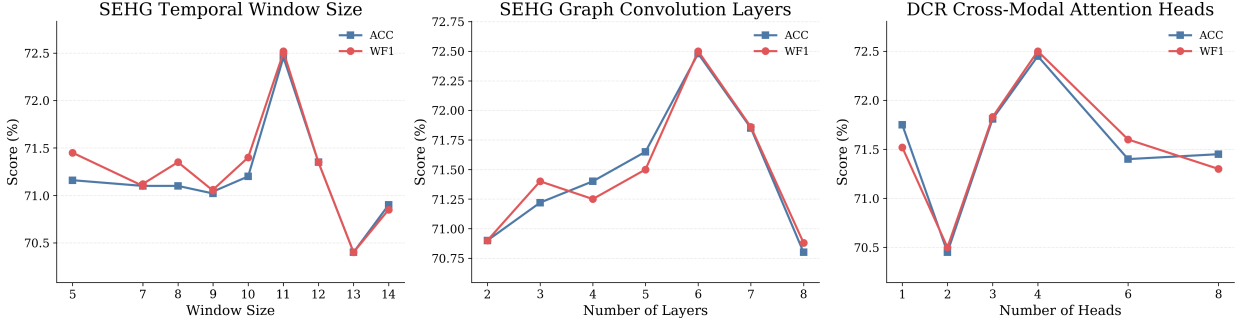


Fig. 9 Parameter sensitivity analysis on IEMOCAP. The left panel shows the effect of the temporal window size W , while the remaining panels show the effect of graph depth L and attention heads H . Overall, the figure reflects the balance between context coverage, structural selectivity, and modeling depth.

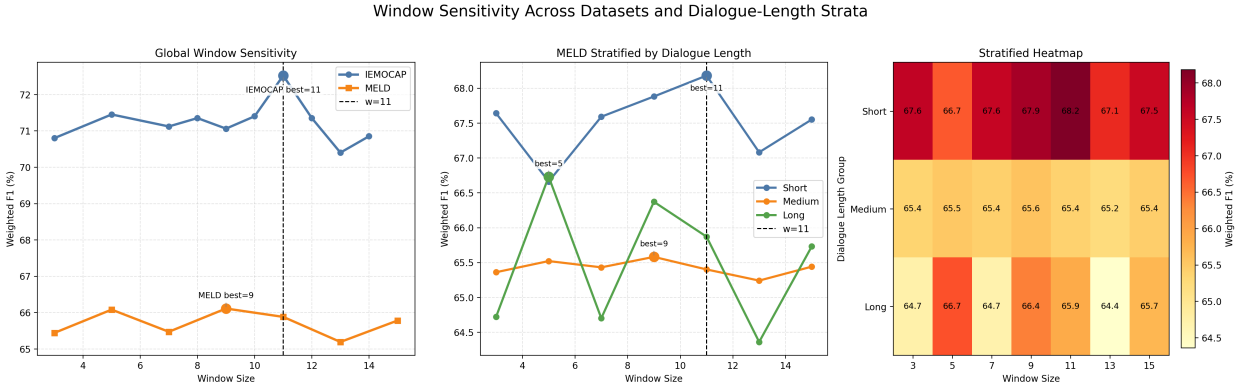


Fig. 10 Extended window-sensitivity evidence across datasets and dialogue-length strata. The left panel compares the global window sensitivity on IEMOCAP and MELD, showing that IEMOCAP is best at $W = 11$ while MELD reaches its corpus-level optimum at $W = 9$. The middle and right panels further stratify MELD by dialogue length, revealing that the preferred receptive range varies across Short, Medium, and Long conversations. These results support our interpretation of $W = 11$ as a robust unified operating point rather than a universally optimal constant.

(P_{in}) via Affective Transition Learning to capture fine-grained emotional turns, producing a logically consistent self-evolution trajectory (P_{out}). This collaborative workflow demonstrates GraphDREAM’s robustness in modeling emotional self-consistency through purified representation, evolutionary topology, and sequential calibration.

5 Conclusion and Future Work

In this work, we propose GraphDREAM, a progressive framework addressing modality heterogeneity, complex topologies, and affective drift in multimodal emotion recognition. By synergizing Decoupled Cross-modal Refinement, Spatio-temporal Evolutionary Heterogeneous Graph, and Multi-task Consistency Calibration, our model establishes a *purification-driven topology and topology-guided calibration* paradigm. GraphDREAM is organized as a coordinated system rather than a loose combination of independent modules: DCR purifies the cross-modal representation space, SEHG reasons over speaker-aware evolutionary topology, and MCC calibrates the resulting sequence into logically coherent emotional trajectories. Extensive experiments on four benchmarks support the effectiveness of this design. Overall, the results suggest that GraphDREAM helps reduce cross-modal redundancy, capture inter-speaker dynamics via evolutionary windows, and alleviate prediction volatility through Affective Transition Learning, providing an interpretable framework for diverse conversational settings.

Limitations

This study also has several limitations. First, the current SEHG uses a fixed symmetric temporal window, which offers a stable default across datasets but may not be optimal for all dialogue lengths or interaction densities. Second, the default graph topology relies on static speaker-aware and temporal edges; although this choice is empirically robust and interpretable in our pilot study, more reliable integration of dynamic implicit edges remains unexplored. Third, most mechanism-oriented ablations and topology controls are centered on IEMOCAP, so their generality to other datasets

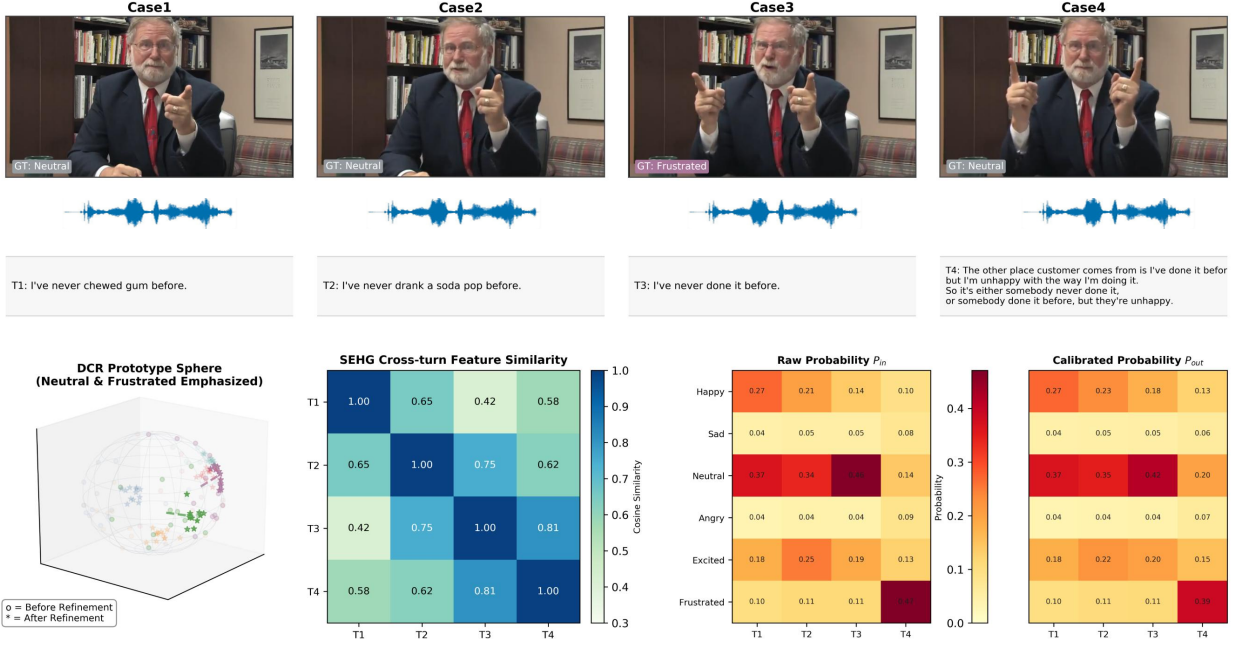


Fig. 11 Mechanism-oriented case study of GraphDREAM on a single-speaker monologue across Case 1 to Case 4. The bottom panels illustrate how DCR enhances the clustering of emotional prototypes in the left subfigure, SEHG aggregates historical context in the center, and MCC ensures a logically consistent self-evolution trajectory P_{out} via probability calibration in the right.

should be interpreted with appropriate caution. Fourth, our refinement-stage analysis focuses on two direct redundancy indicators before graph propagation; additional evidence for representation purification may be explored in future work.

Future Work

Several directions therefore warrant further exploration. Adaptive window mechanisms could automatically adjust the receptive temporal range according to dialogue density and speaker dynamics, instead of relying on a fixed window across all scenarios. Static speaker-aware topology may also be integrated with more reliable dynamic implicit edges so as to preserve interpretability while exploiting high-confidence emotional similarity. Beyond sliding windows, memory-augmented mechanisms could be investigated to capture ultra-long-range dependencies while avoiding over-smoothing in massive histories. Finally, incorporating causal intervention to disentangle genuine emotional triggers from noisy conversational backgrounds may further improve robustness under distribution shift. We hope these directions will help extend the current framework in future research.

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Declarations of Competing Interests

The authors declare no competing interests.

Data Availability

The datasets used in this study are publicly available.

Author Contributions

Yue Wu developed the core algorithms and code, conducted the extensive experiments, analyzed the data, and drafted the original manuscript. **RuPing Hu** and **Yuchun Li** contributed to the data collection, performed part of the comparative

experiments, and assisted in document preparation. **Zhixin Luo** and **Linjie Pan** participated in the literature review, assisted with multimodal representation learning experiments, and visualization of results. **Lingxi Peng** provided the research conceptualization, secured the necessary resources, supervised the entire project, and performed critical revisions of the manuscript. All authors have read and agreed to the published version of the manuscript.

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